



# Universal Design for Learning and Technology

to Support Diverse Learners

LORI COONEY



ICI  
UMASS BOSTON



Lori\_Cooney



Tell me and I  
**forget.** Teach me  
and I **remember.**  
Involve me and I  
**learn.**

~ Benjamin Franklin



# Webinar Overview



Polls



Trends



Activities



Q & A



<https://www.youtube.com/embed/3LDI1QXM7F0>

# TRENDS IN EDUCATION

- **Mobile Devices**
- **Digital Content**
- **Personalized Learning**
- **Online Assessments**
- **Gamification**



Universal Design  
with Technology

# Universal Design **Overview**



# Access through Architectural Design



# Universal Design

**in our daily  
lives**



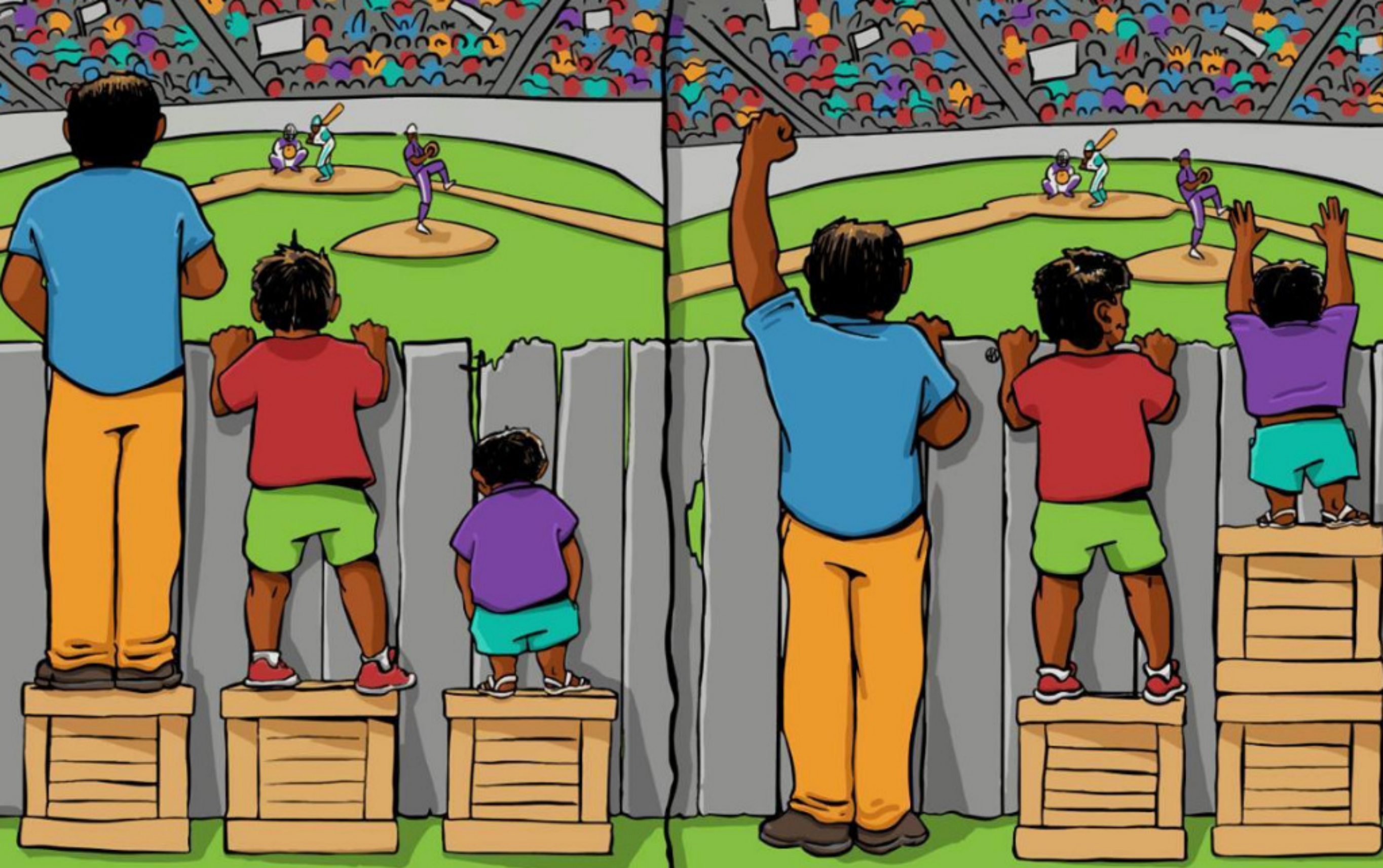


# Universal Design

in our daily  
lives



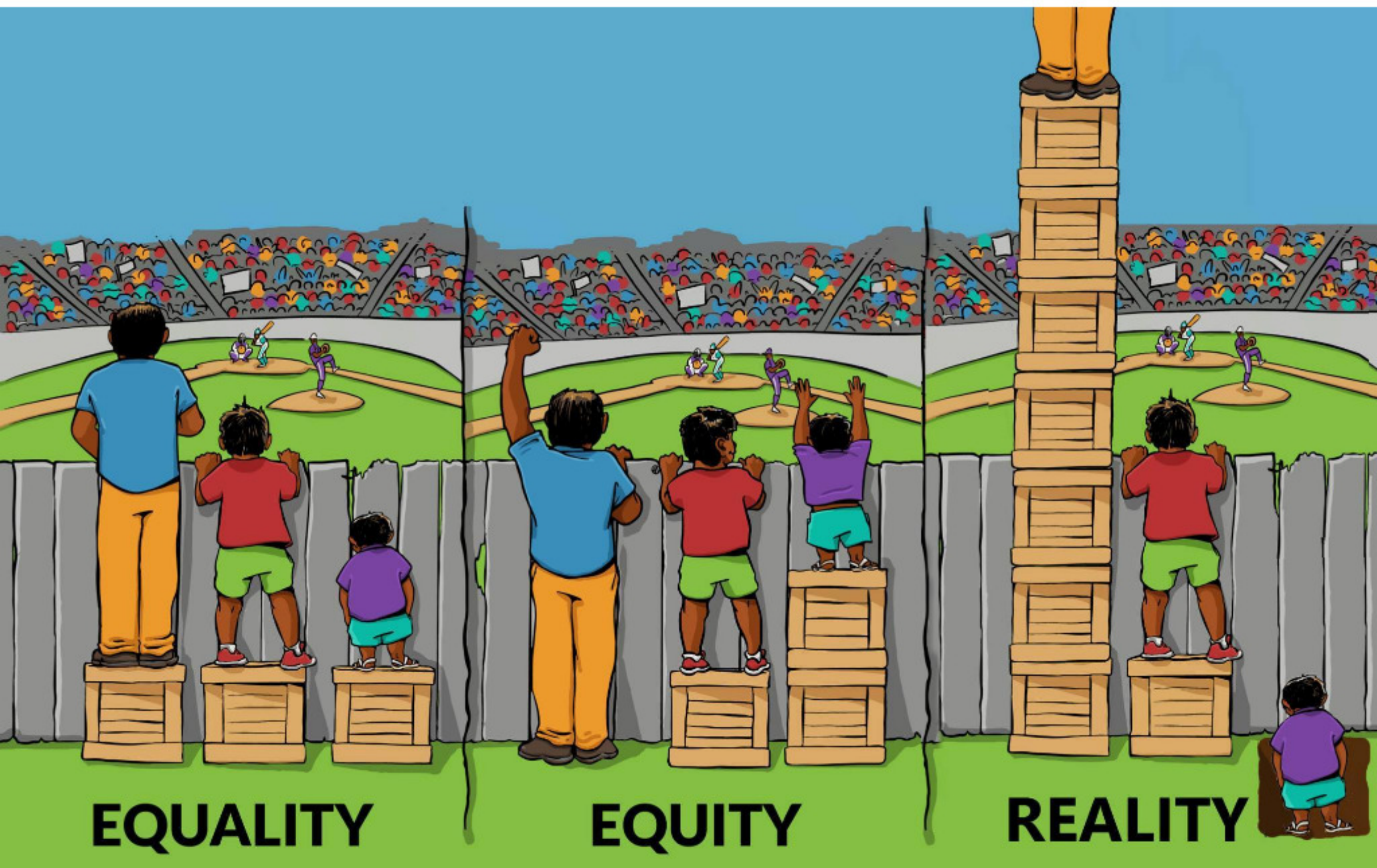




**EQUALITY**

**EQUITY**







# Consider ALL needs from the beginning





# Universal Design for Learning Guidelines

## I. Provide Multiple Means of Representation

### 1. Provide options for perception

- Options that customize the display of information
- Options that provide alternatives for auditory information
- Options that provide alternatives for visual information

### 2. Provide options for language and symbols

- Options that define vocabulary and symbols
- Options that clarify syntax and structure
- Options for decoding text or mathematical notation
- Options that promote cross-linguistic understanding
- Options that illustrate key concepts non-linguistically

### 3. Provide options for comprehension

- Options that provide or activate background knowledge
- Options that highlight critical features, big ideas, and relationships
- Options that guide information processing
- Options that support memory and transfer

## II. Provide Multiple Means of Action and Expression

### 4. Provide options for physical action

- Options in the mode of physical response
- Options in the means of navigation
- Options for accessing tools and assistive technologies

### 5. Provide options for expressive skills and fluency

- Options in the media for communication
- Options in the tools for composition and problem solving
- Options in the scaffolds for practice and performance

### 6. Provide options for executive functions

- Options that guide effective goal-setting
- Options that support planning and strategy development
- Options that facilitate managing information and resources
- Options that enhance capacity for monitoring progress

## III. Provide Multiple Means of Engagement

### 7. Provide options for recruiting interest

- Options that increase individual choice and autonomy
- Options that enhance relevance, value, and authenticity
- Options that reduce threats and distractions

### 8. Provide options for sustaining effort and persistence

- Options that heighten salience of goals and objectives
- Options that vary levels of challenge and support
- Options that foster collaboration and communication
- Options that increase mastery-oriented feedback

### 9. Provide options for self-regulation

- Options that guide personal goal-setting and expectations
- Options that scaffold coping skills and strategies
- Options that develop self-assessment and reflection

# Executive Function

EF is a set of mental processes that helps connect past experiences with present action.



Planning



Organizing



Strategizing



Paying  
Attention



Remembering  
Details



Managing  
Time





Planning  
Projects



Choice  
Activities



Time  
Management



Initiating  
Tasks



Generating  
Ideas

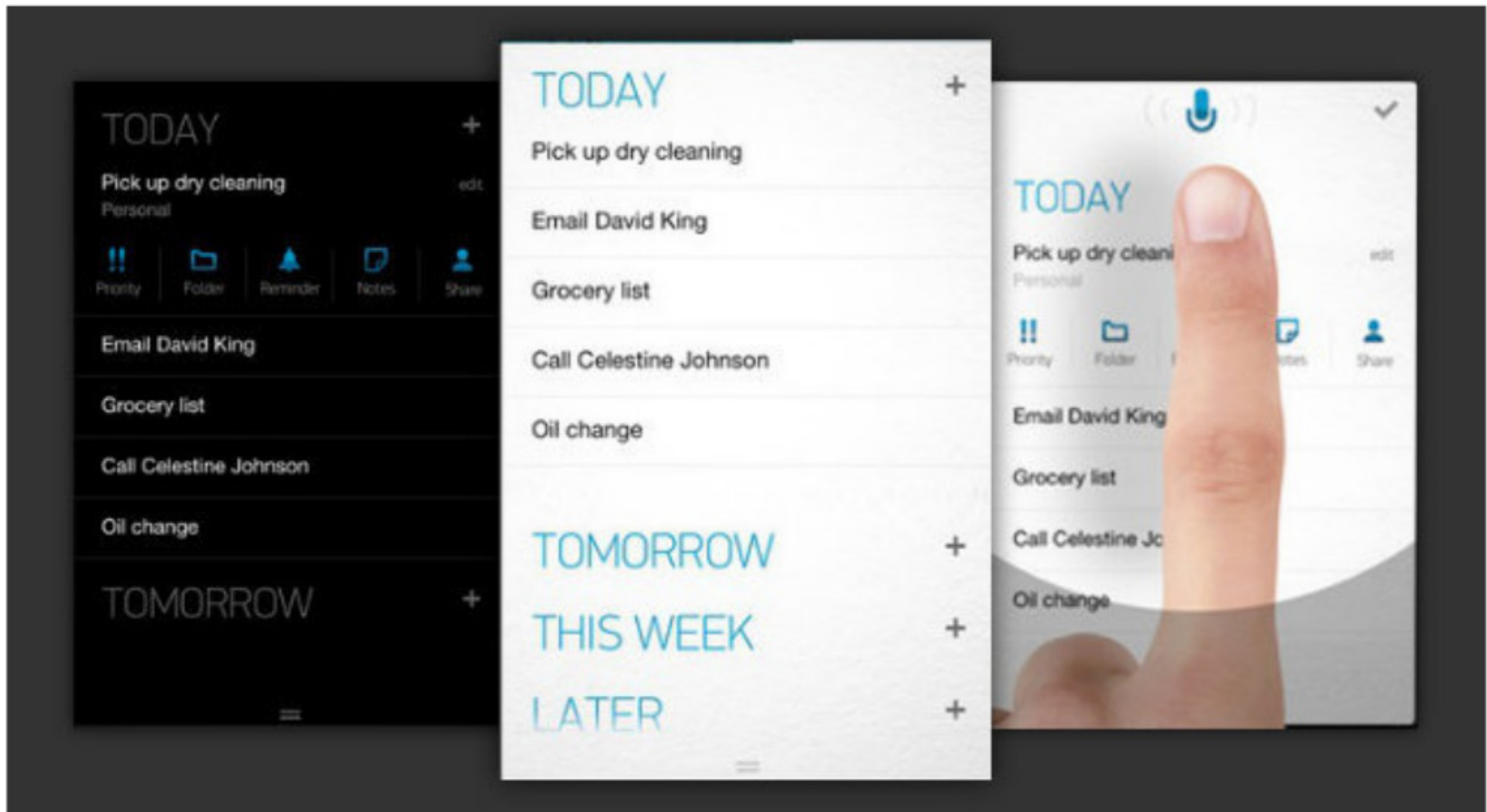


Retaining  
Information

## Executive Function in Education



# Any.do





## Current Trends

# **Differentiated Instruction**



"Build lessons, develop teaching materials, and vary your approach so that all students, regardless of where they are starting from, can learn content effectively, according to their needs."

~Edutopia



Teaching Strategy:

# **Universal Design**

**with Technology**







Multiple Means of  
**EXPRESSION**  
using Technology



Executive Function

Assistive Technology

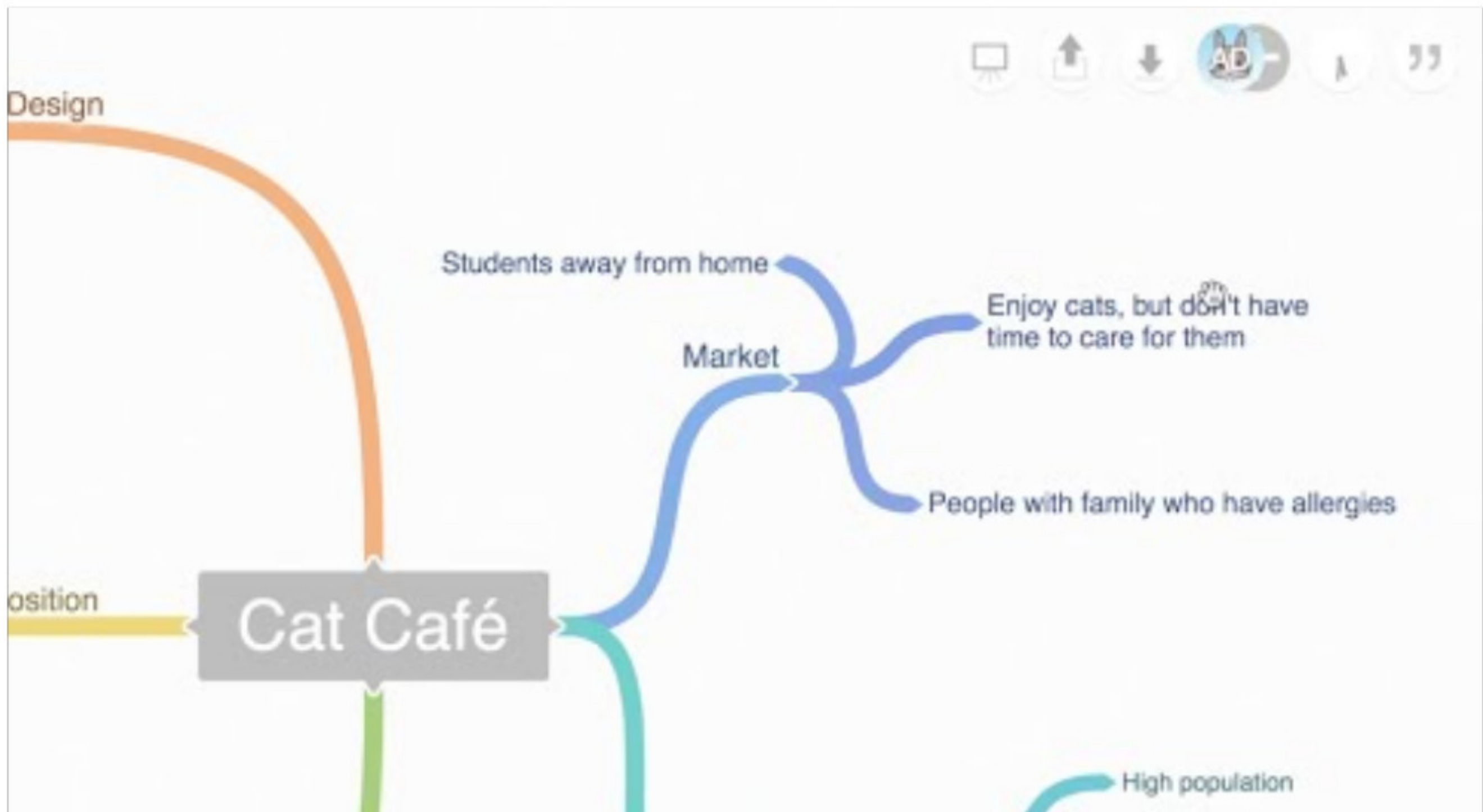
Web Applications

Concept Mapping

Animations

Presentations

# Coggle.it





## Instructional Strategy

Students summarize the most important points of a topic

UDL Suggested activities:



- Draw a picture or cartoon
- Record your summary
- Online or face-to-face class discussion
- Write a 2-minute paper

# Draw It



Draw a picture of yourself as a superhero. Make sure to write down 2-3 words to describe your superhero self.

# Videolicious



The screenshot shows the Videolicious website interface. At the top left is the 'Videolicious' logo in a white script font. To its right are navigation links: 'HOW IT WORKS', 'EXAMPLES', and 'PRICING'. Further right are two green buttons labeled 'SIGN IN' and 'SIGN UP' with a right-pointing arrow. The main content area has a dark background. On the left, the text 'Make great videos! Just talk and tap.' is displayed in white, followed by a smaller line of text: 'Fill your Twitter stream and Facebook page with powerful, professional-quality videos you make yourself—automatically—in seconds.' Below this is a green button labeled 'TRY IT FREE' with a right-pointing arrow. On the right side of the main area is a large image of a hand holding a smartphone, which displays a video player interface with a large play button. Below the main content area, there is a row of three white boxes. The first box contains an image of a smartphone, a tablet, and a laptop, all displaying the Videolicious interface. The second box contains a green 3D bar chart with an upward-pointing arrow. The third box contains a grid of many small, colorful video thumbnails.





[https://www.youtube.com/embed/Ox\\_fG547528](https://www.youtube.com/embed/Ox_fG547528)



Multiple Means of  
**ENGAGEMENT**  
using Technology

Clear Expectations

Vary Instruction

Involve All Students

Presentation Tools

Assessments

Multi-Media



# Poll



**Do you use any of these online assessment tools with your students? (check all that apply)**



Nearpod



Socrative



Google Forms



Kahoot.it



PlayPostIt or another Video Interactive



Other



# NEARPOD FEATURES

- Works on ALL devices
- Multiple feedback options
- MultiMedia (images, audio, text, video)
- Interactive and fun!
- Synchronous or Asynchronous

Accessible: iOS, PDF Viewer for text to speech



## What Is Nearpod?

56,546 views

96 15 SHARE + ...



Nearpod

SUBSCRIBE 1.5K

[https://www.youtube.com/watch?v=g3GVkM\\_GVa4](https://www.youtube.com/watch?v=g3GVkM_GVa4)

Nearpod's free, customizable and ready-to-teach interactive lessons for all K-12 grades and subjects. Engage students one-to-one by synchronizing your lessons across all student

SHOW MORE

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## Up next

AUTOPLAY ☒



### How to create a Nearpod...

Maria Cosio  
3.1K views



### Tutorial "Como utilizar..."

Dulce Ivonne Garcia  
12K views



### I Adopted Rich People's Habits,...

BRIGHT SIDE  
4.5M views



### Grading Hacks #1 for Teachers,...

Laura Randazzo  
27K views



### Google Docs and Sheets Tutorial...



### DITL: Zero Waste Vegan Family!

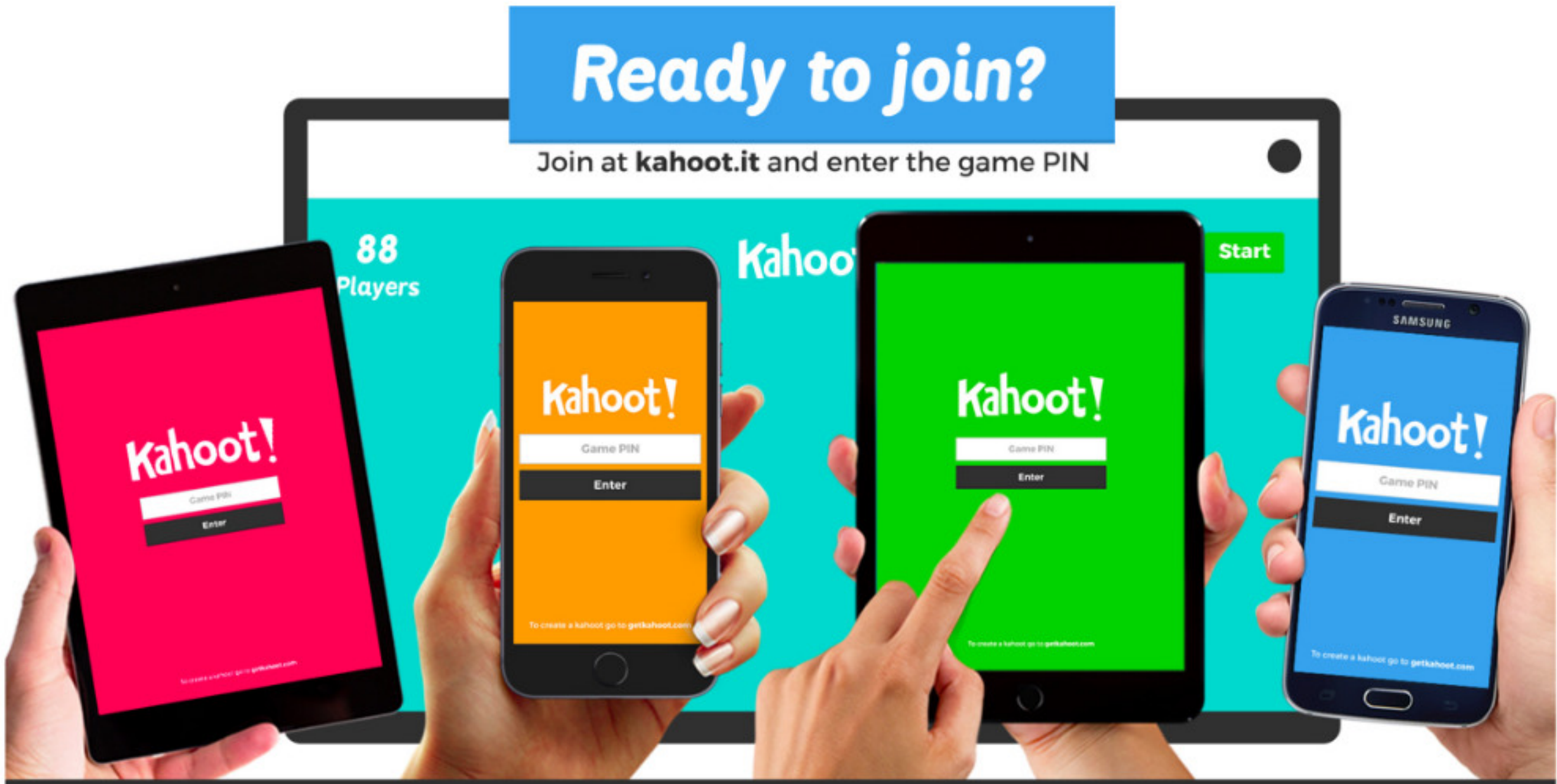
The Fairly Local Veg  
Recommended for you

NINJA

Draw My Life :



# Formative







Kahoot! for businesses: make training and presentations fun!

38,202 views

42 8 SHARE SAVE ...



Kahoot!

SUBSCRIBE 9.6K

<https://www.youtube.com/watch?v=KQRpKhZkVt0&feature=youtu.be>

how Kahoot! is being played in work settings by teams in different companies.

SHOW MORE

6 Comments

SORT BY

Up next

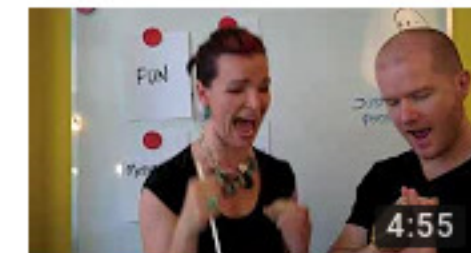
AUTOPLAY



5 Minute Guide to Kahoot  
Tom Driscoll  
142K views



6 Public Speaking Tips T...  
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What Are Your Salary...  
Deniz Sasal  
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3 Jokes That Make People...



Rocklin HS Instrumental...  
Rocklin High Music  
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## CELL STRAINER

from 1  $\mu\text{m}$  to 500  $\mu\text{m}$



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See all   
38 References

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## Kahoot! It: Gamification in H

Article (PDF Available) in [Pertanika Journal of Social Science and Humanities](#) 26(1):565-582 · March 2018 with 2,416 Reads

 Cite this publication



**Debbita Tan**

9.68 · Universiti Sains Malaysia



**Malini G**

32.33 · Universiti Sains Malaysia



Ad



Hosting  
Solutions

[https://www.researchgate.net/publication/320182671\\_Kahoot\\_It\\_Gamification\\_in\\_Higher\\_Education](https://www.researchgate.net/publication/320182671_Kahoot_It_Gamification_in_Higher_Education)

Abstract

Play is a social-centred process, able to boost motivation and promote learning across all levels and ages. With the growing push for creativity in the classroom as well as the

• High Availability  
Infrastructure

Learn More





Please enter the room number:

Room number:

Join Room

*Students open the app or the web address, add the room number and tap "Join Room"*

Quick Quiz

Room number

How many bits in a byte?

2

4

6

8

10

*Questions appear for the students to answer*



Student Login

Room Name



 English ▾

<https://b.socrative.com/login/student/>

Socrative Student Response



**Lori Cooney**  
Search Code: j13cc2



**7**  
VIDEO BULBS



**1**  
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# Please login to access

## Single sign-on



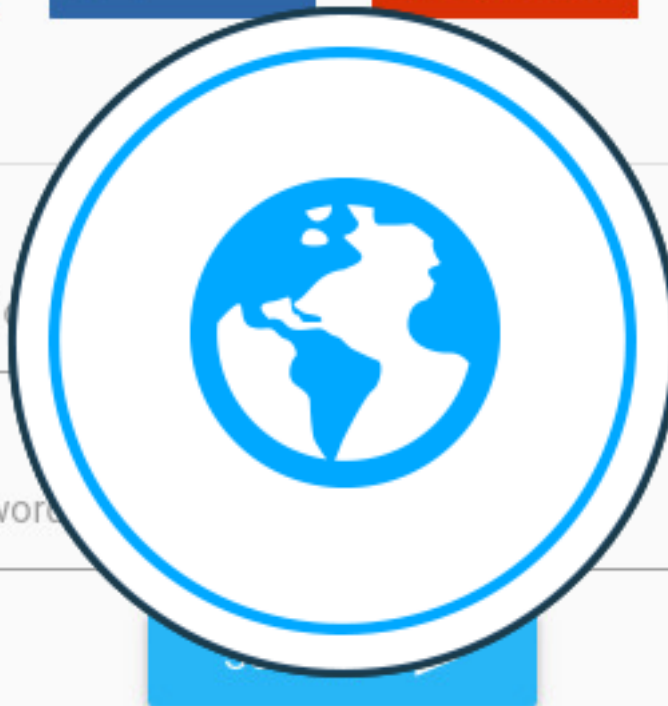
## Login



email



password



<https://www.playposit.com/play/812903/copy-of-note-making>



## What we do

Welcome to loricooney551's presentation!

Introduce yourself

Enter the screen name you would like to appear alongside your responses.

Name



Using a screen name, participants can change your responses. You

<https://pollev.com/loricooney551>

Multiple Means of  
**REPRESENTATION**  
using Technology

Sensory Disabilities

Scaffolding

Step-by-Step Instructions

Outline Information

Digital Imagery

Text-to-Speech



# Canva.com

Education Infographic Templates

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- Book Covers
- Certificates

**HIGHER EDUCATION IN SAN DIAS**

**LIFECYCLE OF A BUTTERFLY**

**STAGE 1: BUTTERFLY EGGS**

**STAGE 2: CATERPILLAR**

**STAGE 3: CHRYSALIS**

**SIMPLE STEPS TO BICYCLE SAFETY**

Make your safety a priority

- 1. PROTECT YOUR HEAD**
- 2. OBEY TRAFFIC LIGHTS**
- 3. SEE AND BE SEEN**
- 4. WATCH FOR ROAD HAZARDS**
- 5. AVOID RIDING AT NIGHT**
- 6. USE HAND SIGNALS**

**CREATING YOUR BUSINESS PLAN**

Business plans involve decisions.

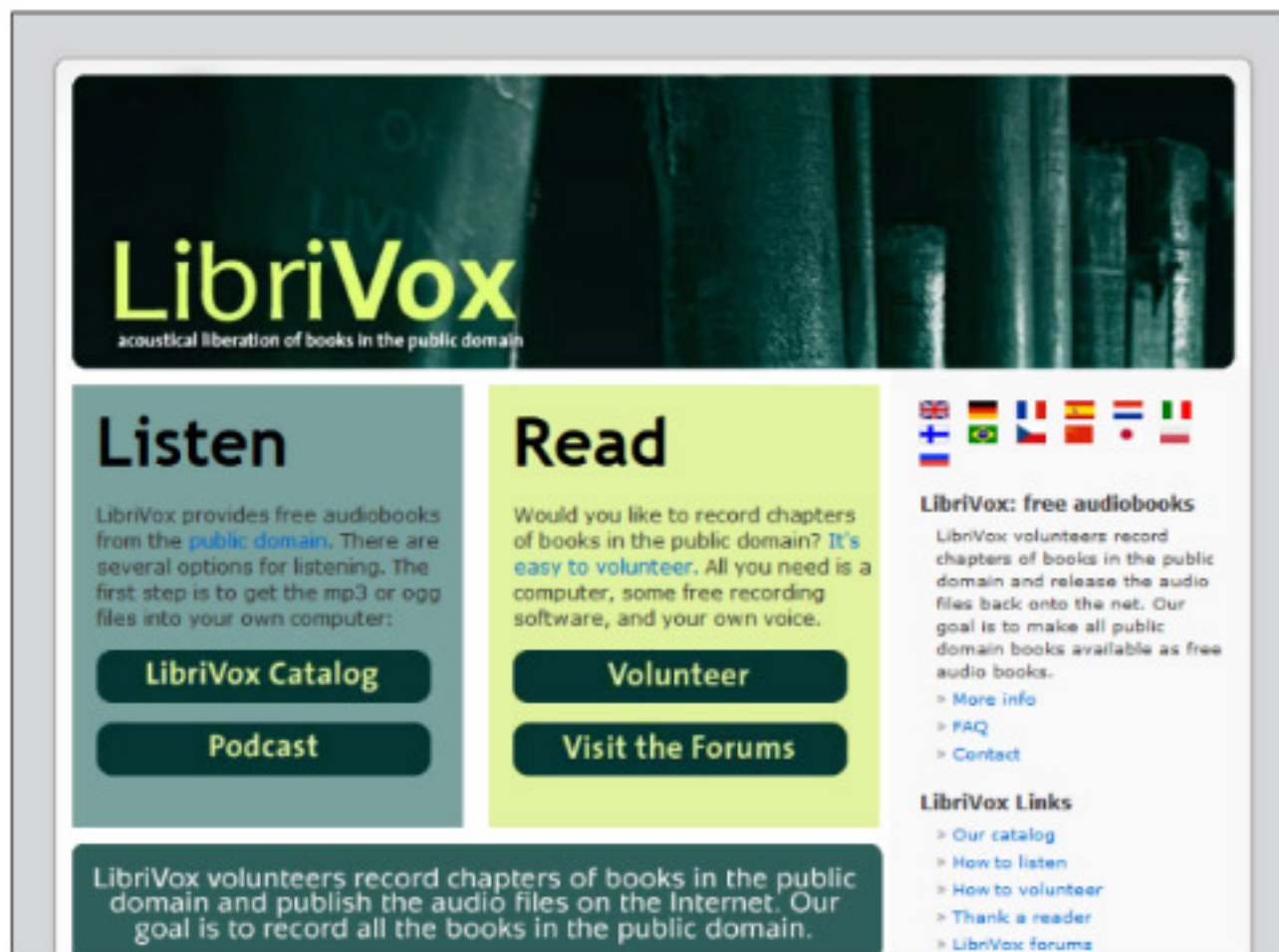
**STARTING A BUSINESS**

**MANAGING YOUR BUSINESS**

**GROWING YOUR BUSINESS**

**EXITING YOUR BUSINESS**

# Librivox + Gutenberg





# Librivox + Gutenberg

VI.

HOPE.

Hope is the thing with feathers  
That perches in the soul,  
And sings the tune without the words,  
And never stops at all,

And sweetest in the gale is heard;  
And sore must be the storm  
That could abash the little bird  
That kept so many warm.

I've heard it in the chillest land,  
And on the strangest sea;  
Yet, never, in extremity,  
It asked a crumb of me.

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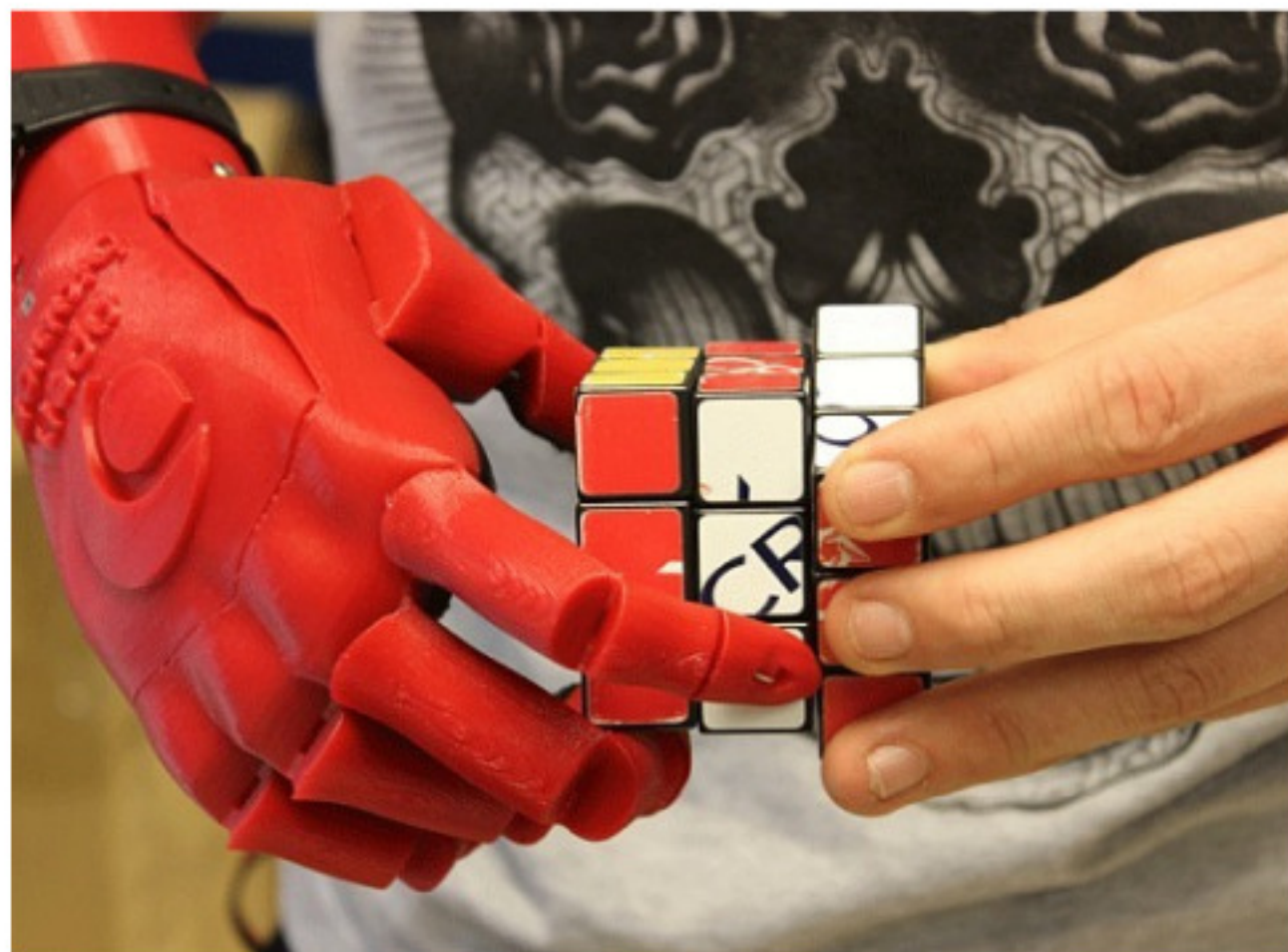
00:00



# Speak it! Google Chrome



# 3d Printing





# Diagramcenter.org

## Building New Paths To Accessibility

Supporting different learning needs with emerging technologies  
and community engagement.

### Our Work



Image  
Description



3D Printing,  
Tactiles & Haptics



Accessible  
Math



Born Accessible  
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Helping build content in  
multiple formats

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Supporting their children's  
ability to succeed

Who We Are



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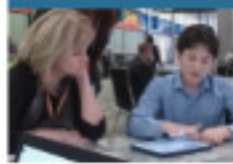
Bookshare offers special summer reading lists for all ages so you can read your way across the U.S. of A.

**633,791 titles  
and counting!**

#### Summer Reading Lists



#### Partnering with Public Libraries



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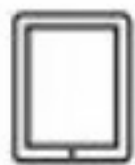
This project is supported by the U.S. Department of Education, Office of Special Education Programs (OSEP).

# Bookshare Read 2 Go

## App



Read2Go



iPad



iPhone



iPod Touch

- Listen with text-to-speech and adjust speed
- Read words are highlighted
- Adjust font size
- Choose colors
- Reads in Braille
- \$19.99



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<https://www.dcmp.org/>

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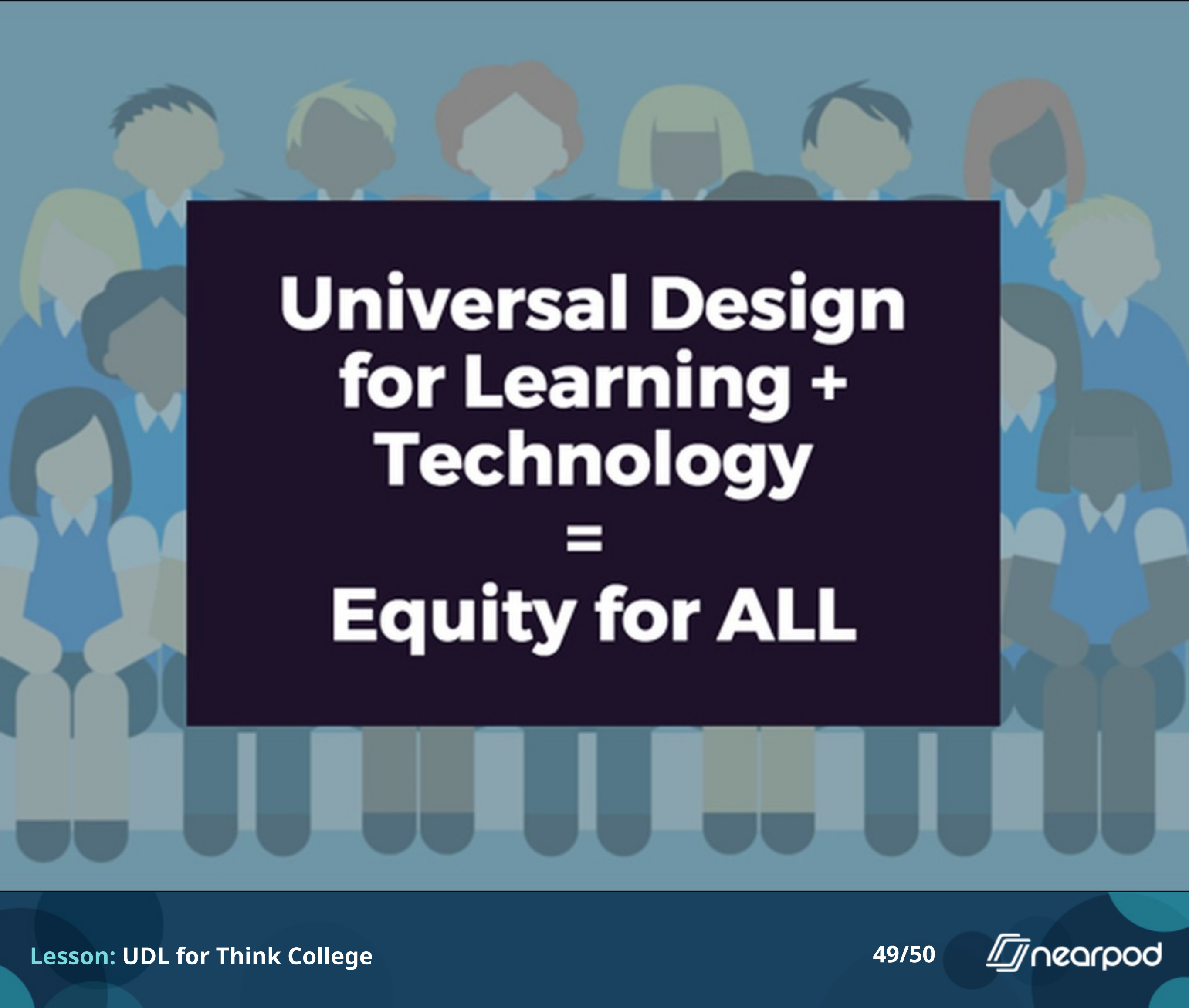


**Plum Island Estuary: Studying How  
Marshes Respond To Sea-Level Rise -  
2009 - 4 minutes**



The estuary is part of the Plum Island Ecosystems LTER



A large group of stylized, semi-transparent student avatars in school uniforms are arranged behind a large, dark purple rectangular sign. The sign contains the text "Universal Design for Learning + Technology = Equity for ALL" in white, bold, sans-serif font. The background is a light blue gradient.

**Universal Design  
for Learning +  
Technology  
=  
Equity for ALL**



# The End!

lori.cooney@umb.edu

THANK ✨ YOU



**Lori\_Cooney**